

Santosh Chaluvvaraju

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- AI Engineer with 6+ years of work experience having robust leadership skills and in-depth AI system comprehension.
- Proficient in Exploratory Data Analysis, Big Data, Statistical Modeling and, crafting scalable end-to-end AI pipelines, Machine Learning Operations from pure research to minimum viable products to scalable production ready deployments.

SKILLS

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|-------------------|-------------------|-----------------|-----------------------|--------------|
| ▪ Python | ▪ DynamoDB | ▪ AWS Sagemaker | ▪ LangChain | ▪ Jenkins |
| ▪ R | ▪ Statistics | ▪ S3 | ▪ Feature Engineering | ▪ REST API |
| ▪ C/C++ | ▪ SVM | ▪ EC2 | ▪ Tensorflow | ▪ Git, Jira |
| ▪ Java | ▪ PCA | ▪ ECS | ▪ Scikit-Learn | ▪ CircleCI |
| ▪ Numpy | ▪ Random Forest | ▪ AWS Lambda | ▪ Transformers | ▪ Pytorch |
| ▪ Data Structures | ▪ XGBoost | ▪ GLUE | ▪ CNN, RNN, LSTM | ▪ FAST API |
| ▪ Algorithms | ▪ Decision Trees | ▪ EDA | ▪ CI/CD | ▪ Flask |
| ▪ SQL | ▪ Neural Networks | ▪ ETL Pipeline | ▪ Docker | ▪ Springboot |
| ▪ MongoDB | ▪ Deep Learning | ▪ Big Data | ▪ Matplotlib | ▪ Kubernetes |

CERTIFICATIONS

- **AWS Certified Machine Learning Speciality** - Amazon Web Services - [Validity: January 2024 - January 2027]
- **AWS Certified Cloud Practitioner** - Amazon Web Services - [Validity: September 2023 - September 2026]
- **Deep Learning Specialization** - DeepLearning.AI
- **Generative AI With Large Language Models** - DeepLearning.AI
- **Exploratory Data Analysis For Machine Learning** - IBM

PROFESSIONAL EXPERIENCE

OBJECT COMPUTING

St. Louis, MO

Machine Learning Engineer - Internship

May 2023 - August 2023

- Leveraged Cloud AI Platform to segment images at scale using Segment Anything Model (SAM) from META.
- Developed Custom prediction routines and containers to deploy SAM to an endpoint for Online Predictions.
- Built Batch Prediction pipelines using Kubeflow to segment images and tested against batch size of up to 1000 images.
- Finally documented and Open Sourced the work in Github repo and also gave a presentation on the usage and impact.
- Designed ML Model workflow flowchart diagrams that represent online prediction and batch prediction pipeline architecture.
- Worked on AWS migration of a CNN model. Built a CI/CD ML workflow pipeline for deploying to the endpoint.

ILLINOIS INSTITUTE OF TECHNOLOGY

Chicago, IL

Graduate Teaching Assistant

January 2023 - May 2023

- Held weekly TA hours to assist students with their coursework, homeworks and project work.
- Held technical sessions on sequence models, Large Language Models and LLM fine tuning using Hugging Face Library.
- Prepared notes and recorded lectures on Spacy, NLTK and tokenization techniques and shared it with students.

WERIZE

Bangalore, India

Machine Learning Engineer

February 2021 - December 2021

- A Finance Tech Company. Built and implemented a tree based Bot model to replicate Credit team's manual underwriting process, resulting in a 60% enhancement in operational efficiency while reducing incremental manpower needs.
- Built a document classification model to automate classification of positive and negative feedback and reviews to further analyze and improve customer support and current loan application process.
- Carried out statistical analysis to understand the necessity of field investigation of customers since it had significant turnaround time, which added additional delay towards loan disbursement.
- Employed Statistical Modeling techniques for Data Extraction, Data Cleaning, Data Screening, Data Exploration, and Data Visualization of structured datasets.
- Setup and managed ETL pipeline using Apache Airflow for data extraction and preparation of structured data.
- Setup and managed batch prediction pipeline for document classification model using batch transform in AWS Sagemaker.
- Integrated Google Auth to the web app, added API filters in backend to grant access to users only within the organization.
- Achieved decoupling across microservices in processing daily EMI collections by integrating AWS Simple Queue Service(SQS).

HIGH PEAK SOFTWARE

Bangalore, India

Software Engineer

August 2018 - February 2021

- Designed and developed server side RESTful web services using Java and Spring Boot web application development frameworks for ADP payroll application for a USA client.
- Involved in requirement understanding and architecting databases using Mysql RDBMS and gave KT to the entire team on the new design and implementation.
- Built a messaging service facilitating seamless client-end user communication, resulting in an 50% efficiency boost in addressing customer doubts and needs.
- Lead a team of five and took self initiatives to resolve blockers and develop ownership consciousness within the team.
- Individually handled a POC project to use Raspberry Pi to display KPI screens on display devices in footwear production units in an intranet setup. Using Raspberry Pi's can reduce the implementation cost by up to 70% including the display device.
- Developed a Multi-threaded python application which can do live video streaming using FFMPEG software suite and collect temperature data and shoot emails on set threshold conditions.
- Deployed 50 raspberry pi's in one of the footwear manufacturing production units as an initial phase of testing in production.

WINIX TECHNOLOGIES

Bangalore, India

Software Engineer

January 2016 - August 2018

- WIP, A comprehensive Mesh Network Technology. Worked on pairing, discovery, and network association parts of WIP.
- Worked on a multithreaded c++ application running on a OpenWRT operating system flashed on MediaTek MT7628A wifi modules and custom development kits from the development partner in Japan.
- Integrated Luci web interface to WIP and implemented device management and firmware upgrade of devices from webpage.
- Actively involved in Bug Fixing and end to end testing of the platform before every release.
- Involved in regular discussions with the team in Japan on process improvements, developing unit test cases and developing a proper release checklist and methodology.
- Worked on a in-house project called WARP (Wireless Advanced Real Time Platform), a Smart IOT Home system. Worked on onboarding google assistant to Raspberry Pi and integrated IFTTT(If this then that) for remote device control.
- Led a team of 3 to participate in a tech exposure event in one of the engineering institutes showcasing existing products and future possible innovations in the IOT domain.

EDUCATION

ILLINOIS INSTITUTE OF TECHNOLOGY

Chicago, IL

Masters of Science in DATA SCIENCE

January 2022 - December 2023

PERSONAL PROJECTS

- **Missing value Treatment using Clustering Technique :** Analysis of Row Based vs Column Based Ignore Methods.
- **Stock Market Gap Phenomenon Analysis :** Stock Market Gap Phenomenon analysis using Statistical Techniques.
- **Scalable End to End Machine Learning Data Pipeline with Kafka Streaming in AWS :** Simplifying MLOps, A Multi-threaded python app capable of handling end to end ML Workflow from data engineering to predictions.
- **Chat Bot, Based on End to End Memory Networks :** Question Answer model trained on Facebook Babi Dataset, Based on the Paper End to End Memory Networks.
- **PEFT fine tuning FLAN-T5 LLM with Reinforcement Learning (Proximal Policy Optimization) :** Fine-tune a FLAN-T5 Large Language model to generate less toxic content using Meta AI's hate speech reward model. Proximal Policy Optimization (PPO) is used to fine-tune and reduce the model's toxicity.
- **Fine-Tune a FLAN-T5 LLM using Performance Efficient Fine-Tuning, Low-Rank Adaptation technique :** Fine-Tune a FLAN-T5 LLM for Enhanced Dialogue Summarization using PEFT Low-Rank Adaptation technique.
- **META Segment Anything Model(SAM) Machine Learning workflow in Google Cloud Platform :** Deploy SAM on GCP using VertexAI for Online Predictions. Built Batch Prediction Pipeline using Kubeflow.